



Digital Forensics

Investigate the Breach of a File Server at a Local Bank

Introduction

The digital forensic investigator plays a most critical role in identifying, secure, and collecting the evidence required for the investigation, as well analyzing them to find the cause for a data breach. Most of the attackers target financial institutions due to high value of customer financial records and personally identifiable information (PII) stored within their systems.

When investigating the breach of a file server at a local bank, the investigator must determine how the breach occurred and at the same time ensure that data is collected and analyzed in accordance with legal requirements and NIST guidelines. Understanding of federal laws that govern computer crime and data protection is essential to conducting a lawful and effective investigation. Also, investigators need to know which data to gather and which techniques to use to acquire data by following NIST standards and maintaining the chain of custody of the evidence.

Question – 1 You have been hired to investigate the breach of a file server at a local bank. The breach resulted in customers' financial records being stolen. Describe two federal laws that would most likely affect this case.

1. Computer Fraud and Abuse Act (CFAA)

Computer Fraud and Abuse Act (CFAA) was enacted by the United States Congress in 1986. This act was designed to address the gap between existing wire and mail fraud laws and growing prevalence of computer crime. Given breach at local bank, the attacker gained unauthorized access to a bank's file server and stole customer financial records, which directly falls under the CFAA.

Specific offences covered by the CFAA include:

- Accessing a computer and obtaining information without authorization, or in excess of authorization.
- Accessing a computer to defraud and obtain value.
- Extortion involving computers.

This law applies to protected computers, which are defined as 'Exclusively for the use of a financial institution or the United States Government, or any other computer, when the conduct constituting the offence affects the computer's use by or for the financial institution or the Government.' Bank systems are considered as protected computers, so this law directly applies to this scenario. The CFAA also covers data theft, fraud, and intentional damage, with penalties ranging from one to ten years depending on the nature of the crime, and higher penalties for repeat offenses and 10 to 20 years of imprisonment.

2. Gramm-Leach-Bliley Act (GLBA)

The Gramm-Leach-Bliley Act (GLBA) provide details of how financial institutions protect and share nonpublic personal information (NPI) of consumers. The GLBA requires banks to have comprehensive security programs to ensure the confidentiality and security of customer records. The data breach resulted in customers' financial records being stolen will affect GLBA Act. According to GLBA Act, the bank has to conduct a investigation and if they find out misuse of information, they have to notify affected customers.

The GLBA Act is primarily regulatory law and assess whether the bank failed to implement adequate security controls. If the bank violate the Act would result in fines and mandatory remediations.

Question – 2

1. Sources use to gather evidence

We can gather evidence from different sources including end point devices like desktop computers, servers, network storage devices, laptops, and following.

- System and security logs include file access logs, authentication logs, and SIEM logs.
- File servers and document repositories.
- Emails and sharing links.
- Network devices logs including Firewalls, IDS/IPS, and proxy servers.
- Access Control Lists.
- Privilege escalations.

2. Types of data encounter for the investigation

i. Volatile Data

Volatile data is Live data, and it will lose soon after power-off the device or reboot the device. This data includes Random Access Memory (RAM), active network connections, running processes, and Log-on sessions.

ii. Non-Volatile Data

Non-Volatile data is Static data, which is stored on physical media, such as Hard Disk Drives (HDD) and Solid-State Drives (SSD). This data includes system logs, deleted files, files and documents, and emails and stored communications.

3. Best handle data acquisition for Volatile and Non-volatile data

i. Volatile Data Acquisition

This includes data stored in RAM, cache, and temporary registers within the CPU. Volatile data needs to be acquired immediately, because power-off the systems will erase the data. We must use live forensic acquisition tools to capture memory, active connections, and running processes. Techniques such as *memory dumping* and the use of volatile data collection tools, like *Volatility* and *Rekall*, are essential in capturing this data effectively. Also, we have to minimize system interaction and fully document all actions taken to maintain the chain of custody of the devices. We can follow *NIST SP 800-86* guidance to preserve evidentiary value during live response.

ii. Non-Volatile Data Acquisition

Non-volatile data remains intact even when power is removed. This includes data stored on hard disk drives (HDDs), SSDs, and external storage media. The acquisition of non-volatile data usually involves creating forensic bit-for-bit images using write blockers of the storage medium while ensuring the data integrity. Use of write blockers prevent any modifications to the original media during acquisition. Tools like *FTK Imager* and *EnCase* are instrumental in facilitating non-volatile data acquisition, offering robust mechanisms to verify the integrity of the acquired data through cryptographic hashing. We can preserve originals and conduct analysis only on verified copies to have baseline. Also, maintaining complete chain of custody of the evidence will be useful when we present evidence for legal formalities.

Conclusion

This lab helps to cover the aspect of digital forensic investigator. Investigating data breaches in financial institutions requires a careful balance between technical expertise, legal compliance, and ethical responsibility. The Computer Fraud and Abuse Act (CFAA) and the Gramm–Leach–Bliley Act (GLBA) provide the legal framework for addressing unauthorized access and the protection of customer financial information and personally identifiable information (PII).

When performing digital forensic investigations, identifying appropriate evidence sources and selecting the correct tools and techniques for data acquisition are key aspects. Also, investigators need to know different methods and values of volatile and non-volatile data acquisition. This lab covers the federal laws involved in financial data breach and best practices of collecting data to perform analysis duties.

References

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